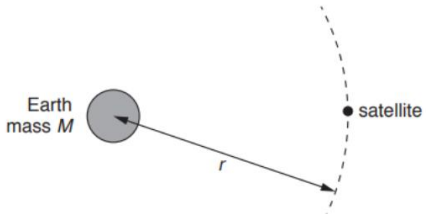
Exercises - Chapter 13 Revision

1. A satellite is in a circular orbit of radius *r* about the Earth of mass *M*, as illustrated in Fig.



The mass of the Earth may be assumed to be concentrated at its centre.

a. Show that the period T of the orbit of the satellite is given by the expression [3]

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

b.

i. A satellite in geostationary orbit appears to remain above the same point on the Earth and has a period of 24 hours.

State two other features of a geostationary orbit.

1.....

2. .....[2]

ii. The mass M of the Earth is 6.0×10^{24} kg.

Use the expression in (a) to determine the radius of a geostationary orbit.

radius=..... m [2]

c. A global positioning system (GPS) satellite orbits the Earth at a height of 2.0×10^4 km above the Earth's surface. The radius of the Earth is 6.4×10^3 km.

Use your answer in (b)(ii) and the expression

$$T^2 \propto r^3$$

to calculate, in hours, the period of the orbit of this satellite.

period=..... hours[2]

[Total: 9]

2.

a.

i. State what is meant by a line of force in a gravitational field.

.....

.....[1]

ii. By reference to the pattern of the lines of gravitational force near to the surface of the Earth, explain why the acceleration of free fall near to the Earth's surface is approximately constant.

.....

.....

.....

.....

.....[3]

b. The Moon may be considered to be a uniform sphere that is isolated in space. It has radius 1.74×10^3 km and mass 7.35×10^{22} kg.

i. Calculate the gravitational field strength at the Moon's surface.

gravitational field strength = ... N kg^{-1} [2]

ii. A satellite is in a circular orbit about the Moon at a height of 320 km above its surface. Calculate the time for the satellite to complete one orbit of the Moon.

time = ... s[3]

[Total: 9]

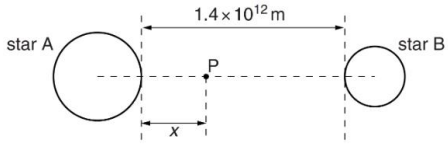
3.

a. By reference to the definition of gravitational potential, explain why gravitational potential is a negative quantity.

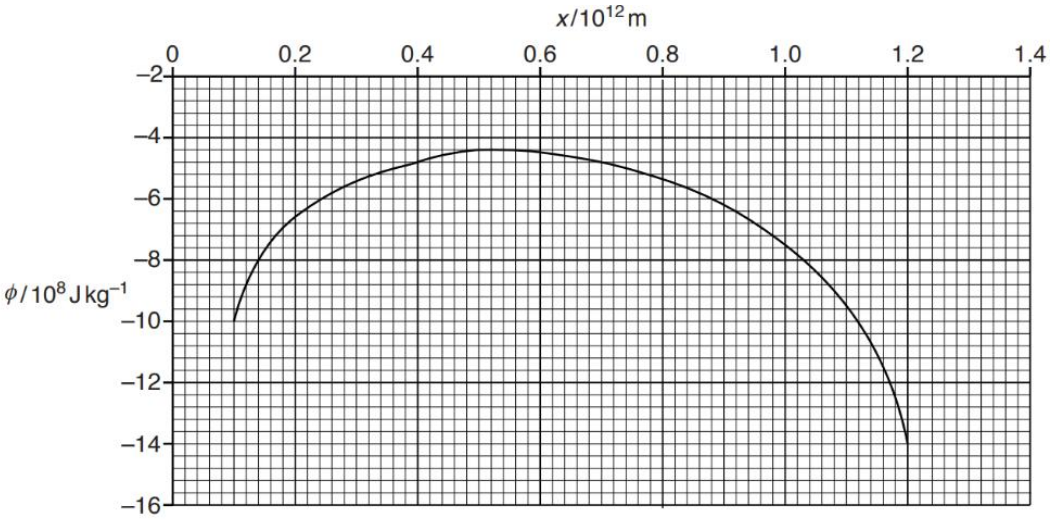
.....

.....[2]

b. Two stars A and B have their surfaces separated by a distance of 1.4×10^{12} m



Point P lies on the line joining the centres of the two stars. The distance x of point P from the surface of star A may be varied.



A rock of mass 180 kg moves along the line joining the centres of the two stars, from star A towards star B.

i. Use data from Fig.1.2 to calculate the change in kinetic energy of the rock when it moves from the point where $x = 0.1 \times 10^{12}$ m to the point where $x = 1.2 \times 10^{12}$ m. State whether this change is an increase or a decrease.

change in kinetic energy = ...J

change:.....[3]

ii. At a point where $x = 0.1 \times 10^{12}$ m, the speed of the rock is v.

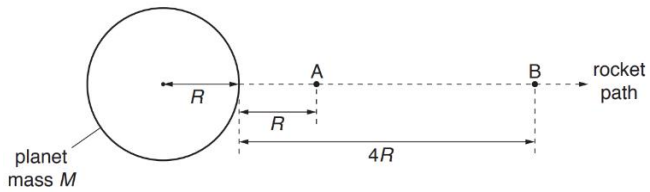
Determine the minimum speed v such that the rock reaches the point where $x=1.2\times10^{12}$ m.

minimum speed = ...m s^{-1} [3]

[Total: 8]

4.

a. A rocket is launched from the surface of a planet and moves along a radial path



The planet may be considered to be an isolated sphere of radius R with all of its mass M concentrated at its centre. Point A is a distance R from the surface of the planet. Point B is a distance 4R from the surface.

i. Show that the difference in gravitational potential Δϕ between points A and B is given by the expression [1]

$$\Delta\phi = \frac{3GM}{10R}$$

ii. The rocket motor is switched off at point A. During the journey from A to B, the rocket has a constant mass of 4.7×10^4 kg and its kinetic energy changes from 1.70TJ to 0.88TJ. For the planet, the product GM is 4.0×10^{14} Nm^2 kg^{-1}. It may be assumed that resistive forces to the motion of the rocket are negligible.

Use the expression in (a)(i) to determine the distance from A to B.

distance = ...m [3]

[Total: 4]